

Project Proposal: Ship Detection in Satellite Imagery

1. Goals

Build a binary classifier that detects ships in satellite image chips. We will train and individually compare traditional ML models on hand-crafted image features against a fine-tuned CNN on raw pixels, and use unsupervised methods to explore the feature space. The central question is whether engineered features can compete with learned representations on this task.

2. Data and Preprocessing

Source: Kaggle “Ships in Satellite Imagery” (kaggle.com/datasets/rhammell/ships-in-satellite-imagery) — 4,000 labeled 80×80 RGB chips (1,000 ship / 3,000 no-ship) from PlanetScope at 3 m resolution.

Preprocessing:

- Normalize pixels to [0, 1]. Apply data augmentation (flips, rotations, brightness jitter) for CNN training to mitigate the small dataset size.
- Extract 25+ tabular features per image for traditional ML: RGB/HSV channel statistics, Canny edge density, GLCM texture descriptors (contrast, entropy, homogeneity, correlation), and Hu moments.
- Stratified 80/20 train/test split to preserve class ratio. Standardize all extracted features.
- Apply class weighting across all models to account for the 1:3 ship-to-background imbalance.

3. Methods

Each model will be trained and evaluated independently, then compared side by side on the same test set to determine which approach performs best.

Unsupervised Models: PCA for dimensionality reduction and feature importance analysis; K-Means clustering to assess whether ship and non-ship images form natural groupings without labels; t-SNE for 2D visualization of cluster structure.

Supervised Models:

- **Baseline** — Logistic Regression with default hyperparameters on the extracted features.
- **Tuned Traditional Models** — Random Forest, SVM (RBF kernel), and XGBoost, each tuned via RandomizedSearchCV with 5-fold cross-validation.
- **CNN (Transfer Learning)** — Fine-tune a pretrained ResNet18 on the augmented image data. We will freeze early layers, replace the classification head, and train with Adam, learning rate scheduling, class-weighted cross-entropy loss, and early stopping on a held-out validation set.

Evaluation: All models compared on accuracy, precision, recall, F1-score, and ROC-AUC. Confusion matrices and ROC curves reported.

Extension: Time permitting, we will re-train the best-performing supervised model on PCA-reduced features to test whether dimensionality reduction improves or degrades classification performance. This ties the unsupervised and supervised analyses together and provides an additional comparison point for the report.

4. Team

Individual project.